Time Series Analysis

China's Export and Import Value by Region



Department of Statistics and Actuarial Science
The University of Hong Kong

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1 Introduction

Import and export, to some extent, indicate a country's (region's) opening up. Since the Chinese economic reform in 1978, China's integration into the global economy has posed a significant effect both locally and globally. Scholars have come to the conclusion that the growth in import and export has significantly benefited the economies, especially in a fast-growing developing country like China [4]. These days, with more and more nations misusing trade emergency procedures, launching investigations into anti-dumping and anti-subsidy of imports, and imposing hefty import duties, paying close attention to the growth of imports and export data of China is crucial for maintaining a sustainable local economic development and designing various internal and external policies. More importantly, China, as the world's second-largest economy in terms of nominal GDP in 2022, demonstrates a vital development benchmark on the global stage. Therefore, It is vital for us to produce an accurate forecast of China's total import and export value over a specific time period in the future.

Though large in economic volume and exerting its influence globally, China has shown specific interests in certain areas over other countries, such as its immediate Western Pacific and Asia Pacific neighborhoods [7]. Therefore, rather than looking at total import and export as a whole, we examine import and export patterns with the Asia-Pacific Economic Cooperation (APEC), the European Union (EU), and the Association of Southeast Asian Nations (ASEAN) respectively. The choice of these three organizations lies in their criticality in political geography and enormous trade values annually.

With the background stated above, this study aims to investigate and predict the trend of China's foreign trade by focusing on the export and import value between China and APEC, EU, and ASEAN, respectively. Previous studies have shown that time series model, such as the Seasonal Autoregressive Integrated Moving Average (SARIMA) model, is useful for analyzing the cyclical behavior of the economy and making predictions, which increases the credibility of our approach to the study [10]. However, in view of many political turmoils and the outbreak of the pandemic, we investigate multivariable time series models, such as vector auto-regression beyond SARIMA. Moreover, to accurately and effectively identify possible external variables, we resort to change-point analysis to detect one more change-points in the mean. Accordingly, we can trace back the time of changes to pinpoint external variables precisely.

This paper unfolds in the following way. In Section 2, we clearly describe the data collection process and show the time plots of the six time series. Section 3 explains Seasonal ARIMA, a univariate model, to forecast imports and exports. In Section 4, we extend the univariate analyses into multivariate ones to account for possible unexplained variances. Change-point analysis, which is leveraged to determine external variables, is lucidly introduced and employed in our time series in Section 5. Finally, Section 6 concludes the paper and proposes some extensions to the current research.

2 Data Collection

We grabbed the data from the official website of China's General Administration of Customs ([1]). In total, we have six time series with three import time series and three export time series with ASEAN, EU, and APEC respectively. As shown in Figure 1, there are 120 monthly data of the Total Export and Import Values with units of US dollars by region from January 2012 to December 2021 for model fitting. Additionally, we reserved six months of data from January 2022 to June 2022 for prediction and model comparison.

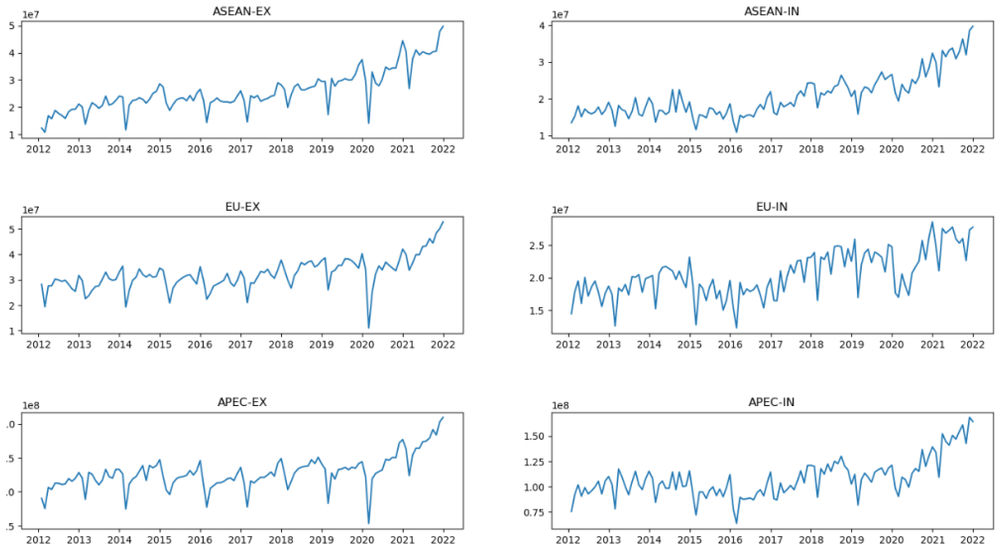


Figure 1: Total Export and Import Values

Before diving into the modeling part, we can gain some insights from the time plots. First, it is easy to observe that all of the six original time series have non-constant variance and mean, hence non-stationary. Therefore, it reminds us to perform log transformation and a common differencing in the modeling process. Second, there is a strong seasonal auto-correlation relationship, indicating that a Seasonal ARIMA model may fit better than a simple ARIMA model. Moreover, the consistent upward trend in all-time plots since 2020 suggests COVID-19 could be one of the external factors we need to account for.

[1] `http://english.customs.gov.cn/`

3 SARIMA Model

In this section, we are going to briefly introduce SARIMA by comparing it with simple ARIMA. Then, we exemplify the model identification, parameter estimation, diagnostic checking, and forecasting using China export to ASEAN countries. In the end, we present the model specifications and forecast results of all six time series.

3.1 Introduction to ARIMA and SARIMA

ARIMA stands for Auto-Regressive Integrated Moving Average, which is widely used in time series. Auto-Regressive indicates that a time series is a linear function of lagged observations. Moving Average means the regression error is a linear combination of error terms whose value occurred contemporaneously and at various times in the past [6]. Integrated refers to the differencing steps are applied one or more times to eliminate the non-stationarity of the trend. In one word, ARIMA assumes the future value is a linear combination of past values and errors.

SARIMA, which stands for Seasonal ARIMA, enhances ARIMA by taking into consideration seasonal variations. Compared with ARIMA models with parameters (p, q)(P, Q), SARIMA is determined by (p, d, q) (P, D, Q, s). The following is the mathematical formulation:

$$W_t = \Phi_1 W_{t-s} + \Phi_2 W_{t-2s} + \cdots + \Phi_P W_{t-Ps} + \phi_1 W_{t-1} + \phi_1 W_{t-2} + \cdots + \phi_p W_{t-p}$$

$$+a_t - \Theta_1 a_{t-s} - \Theta_2 a_{t-2s} - \Theta_Q a_{t-Qs} - \theta_1 a_{t-1} - \theta_2 a_{t-2} - \theta_q a_{t-q}$$

$$\Longleftrightarrow \Phi(B)\phi(B)W_t = \Theta(B)\theta(B)a_t$$

$$W_t = \bigtriangledown^d \bigtriangledown_s^D Z_t = (1-B)^d (1-B^s)^D Z_t$$

where:

- p: Lag parameter in AR model
- d: Number of differencing to generate stationarity
- q: Lag parameter in MA model
- s: seasonal period for a characteristic appearing in a circle
- P: Seasonal autoregressive order.
- D: Seasonal difference order.
- Q: Seasonal moving average order.

3.2 Model specification

In subsections 3.2 to 3.6, we focus on ASEAN-EX to demonstrate our SARIMA model fitting and forecasting process. As indicated in Figure 1, ASEAN-EX is non-stationary. We first took a log transformation. By comparing the ACF plot on logged data before and after the common

differencing, we can observe that ACF decays much faster after differencing as shown in Figure 2. Therefore, we performed a common differencing afterward. Due to the seasonality of data, we further conducted a seasonal differencing as in Figure 3. To be rigorous, we apply the Ljung-box test and the Augmented Dickey-Fuller test to check the problem of over-differencing and under-differencing. Both tests give the positive results that after a common differencing and seasonal differencing on logged data, the processed sequence is stationary and not a White Noise. We determine d = 1, D = 1, and s = 12 due to the monthly data source.

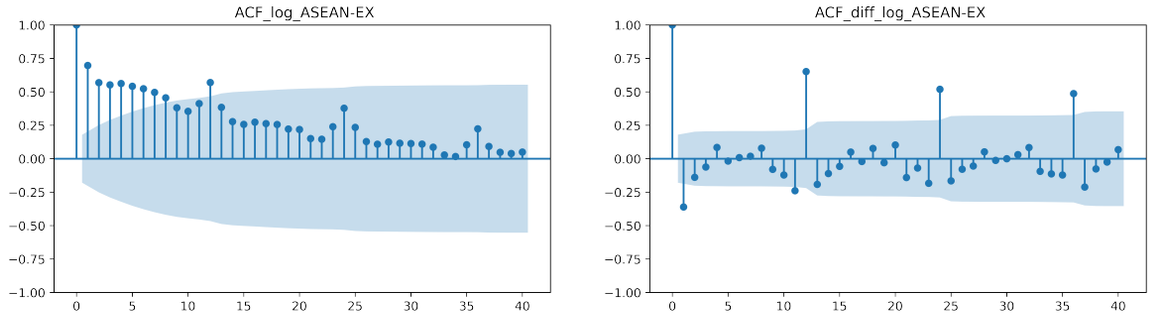


Figure 2: ACF plot of log-data $ln(Z_t)$ and $\bigtriangledown^d ln(Z_t)$ for ASEAN-EX

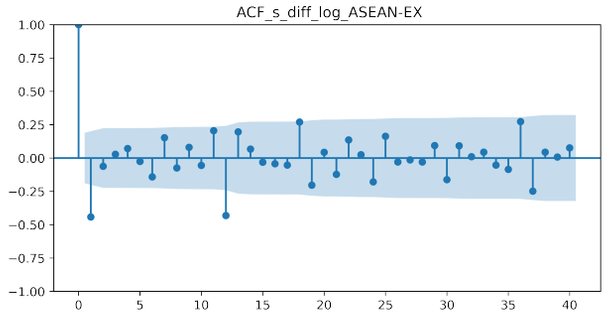


Figure 3: ACF plot of $\bigtriangledown^d \bigtriangledown_s^D ln(Z_t)$ for ASEAN-EX

According to the ACF plot of $\bigtriangledown^d \bigtriangledown_s^D ln(Z_t)$, there is a snapping correlation at lag 1, which decays very fast and close to 0 after. For the seasonal part, the ACF signal stands out at lag 12 and decreases progressively at lag 24, and 36 manifesting itself as relatively more significant than neighboring time periods. We observe an MA(1) $\times$ SAR(1) process. Though at lag 18 auto-correlation also impends over the bound, we dismiss it as a minor feature. The combination of two characteristics leads to the phase-based model of SARIMA (0,1,1) $\times$ (1,1,0,12) on log-transformed data.

3.3 Model diagnostics

After going through the first round of model specification, we conducted model diagnostics through Normality Tests on the residual sequence. The whole residual sequence does not pass the Shapiro-Wilk normality test because the p-value is lower than 0.05, indicating a rejection of the null hypothesis. Meanwhile, the specified model underperformed in Histogram Plot by contrast to KDE(Kernel Density Function) and Normal Density Function. There are more than 10 points deviating from the Q-Q plot on normal distribution, which led us to reject the hypothesis of Normality. Therefore, we concluded that the residual sequence did not follow a normal distribution.

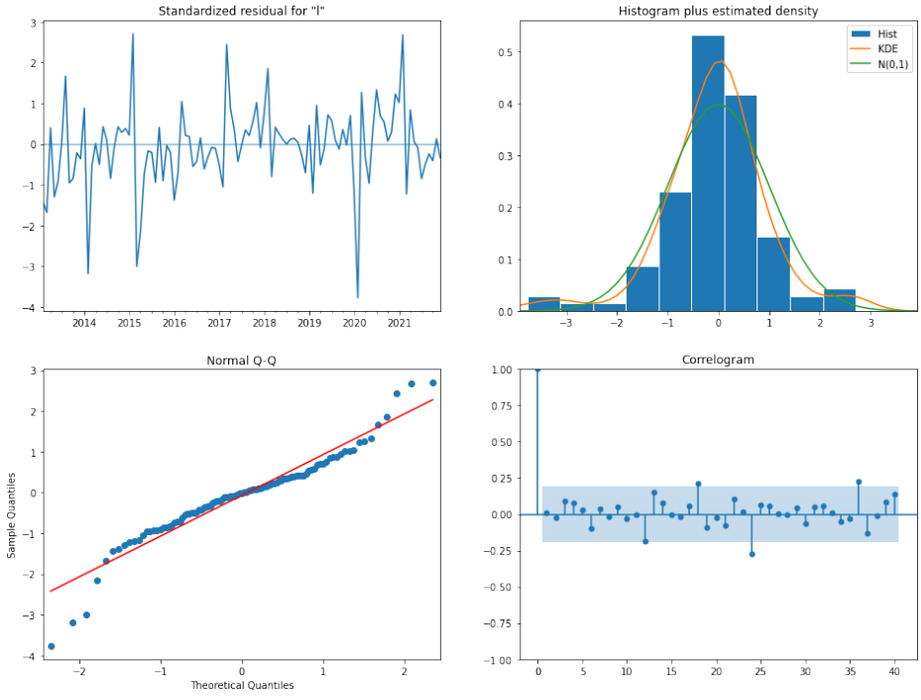


Figure 4: Model Diagnostics for SARIMA(0,1,1)×(1,1,0,12) for ASEAN-EX

Then we examined joint correlations and individual correlations using the Ljung-Box test and the residual ACF plot respectively. We rejected the null hypothesis that the residual is White Noise by using Ljung-Box on the residuals, which shows p-values greater than 0.05 in the first 24 lags and directed us to observe patterns in the residuals. The residual ACF plot in 5 reflects significant correlations at lags 12, 24, 36,..., indicating that our current model is inadequate. Accordingly, we added an SAR(1) process to the current model.

After fitting the enlarged model of (0,1,1)×(2,1,0,12), we repeated the procedure of the preceding model diagnostics(Figure 5). Shapiro-Wilk test on the new sequence gives a higher p-value larger than 0.05, and Histogram Plot is now closer to the theoretical density function. Nevertheless in the Q-Q plot, there are still 4 points relatively isolated from the probability

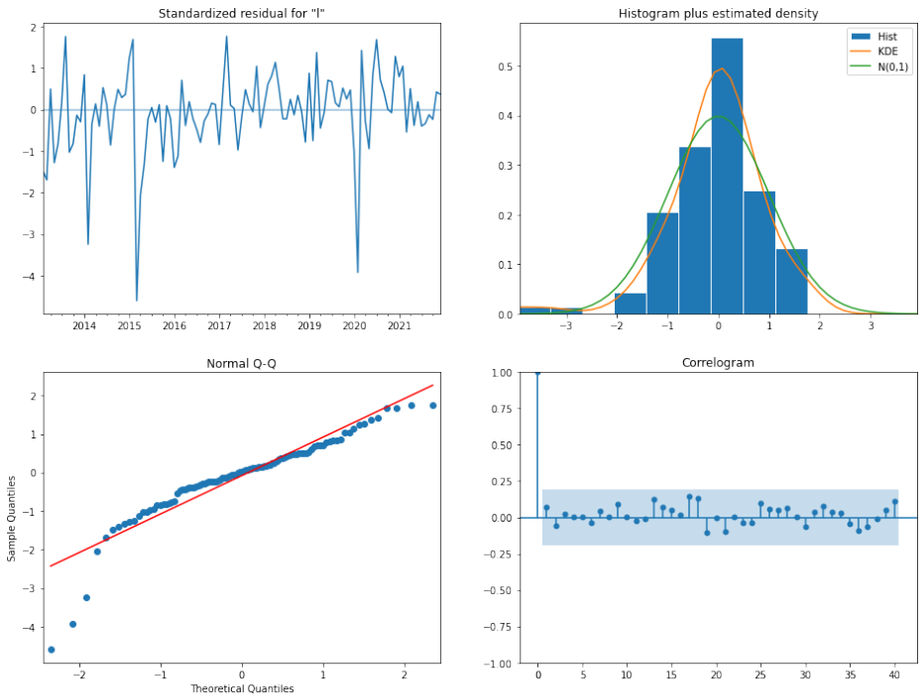


Figure 5: Model Diagnostics for SARIMA(0,1,1)×(2,1,0,12) for ASEAN-EX

line, which corresponds to non-extant information not captured by the SARIMA model. One point is from February 2020, a much sharper dip in actual observations compared to the model estimation. Another 3 points are all from the early stage before 2014 when the model cannot use much historical data, and it makes sense to allow some bias. Meanwhile, the ACFs on the enlarged model are all within the bound, and we cannot recognize a pattern. We finally checked by Ljung-box test and tentatively concluded that the residual sequence can be considered as White noise and stop, and our current model is adequate.

3.4 Model selection

The above modelling yields two SARIMA models:

$$Model\ 1 : SARIMA\,(0,1,1) \times (1,1,0,12)$$

$$Model\ 2 : SARIMA\,(0,1,1) \times (2,1,0,12)$$

We choose the Akaike information criterion (AIC) instead of the Bayesian information criterion (BIC) as the selection criterion as the data size is relatively small at around 120 observations. By comparison, Model 2, the enlarged one, has a lower score which indicates better performance. We design Model 3 by adding another SAR(1) process to Model 2 to illustrate the problem of overfitting if further increase the model size.

model	(0,1,1)×(1,1,0,12)	(0,1,1)×(2,1,0,12)	(0,1,1)×(3,1,0,12)
AIC	-145.35	-166.02	-165.12

Table 1: AIC scores for model selection

3.5 Parameter estimation and interpretation

Table 2 gives a summary of parameter estimation on the best-selected model SARIMA(0,1,1)×(2,1,0,12), and the corresponding time series is represented by:

$$W_t = -0.7452W_{t-12} - 0.4926W_{t-24} + a_t + 0.6389a_{t-1}$$

and $W_t = \bigtriangledown^d \bigtriangledown_s^D ln(Z_t)$ is the data after log transformation and differencing. All the coefficients are statistically significant, with p-values close to 0. The model reduction is unreasonable from hypothesis testing. As illustrated in Figure 6, the model estimation starts from 2013 - 2021. A salient characteristic captured by the model is sharp yearly drops in Februaries with relatively gentle ascends in Januaries and March. This phenomenon could be explained by the number of working days in China, which are closely related to economic production activities. For example, in 2022, China has 21 and 19 working days in January and March, while only 16 in February.

parameter	coef	std err	Z	p-value
MA(1): θ_1	-0.6389	0.067	-9.591	0.000
SAR(1): Φ_1	-0.7452	0.083	-9.026	0.000
SAR(2): Φ_2	-0.4926	0.071	-6.948	0.000
VAR(a_t): σ^2	0.0104	0.001	11.365	0.000

Table 2: Model summary for ASEAN-EX

3.6 Model forecast

Figure 7 provides a contrast of forecast values and real observations from January 2022 to June 2022. Mean Absolute Error (MAE) is introduced to evaluate the performance of SARIMA models with other models in the next section. The MAE of our selected SARIMA model for ASEAN-EX is 0.07, less than one-thousandth of the logged data in the range [17.0, 17.8]. Therefore, we believe the current model provides satisfactory 6-step forecasts.

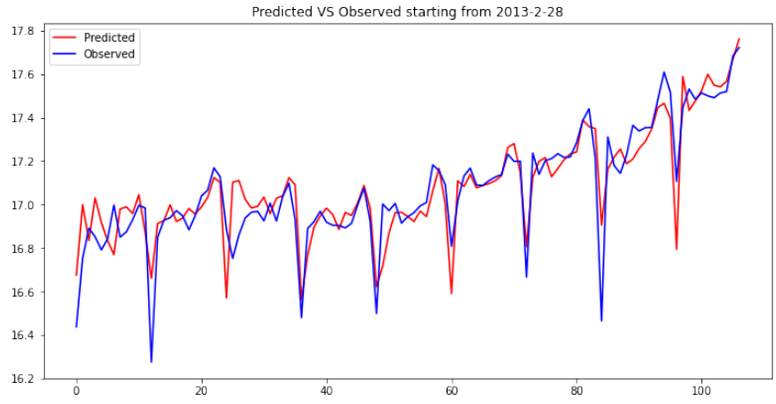


Figure 6: Model prediction: SARIMA(0,1,1)×(2,1,0,12) for ASEAN-EX

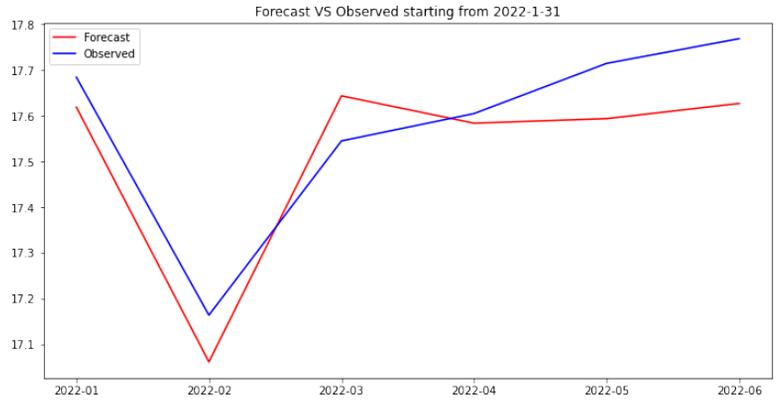
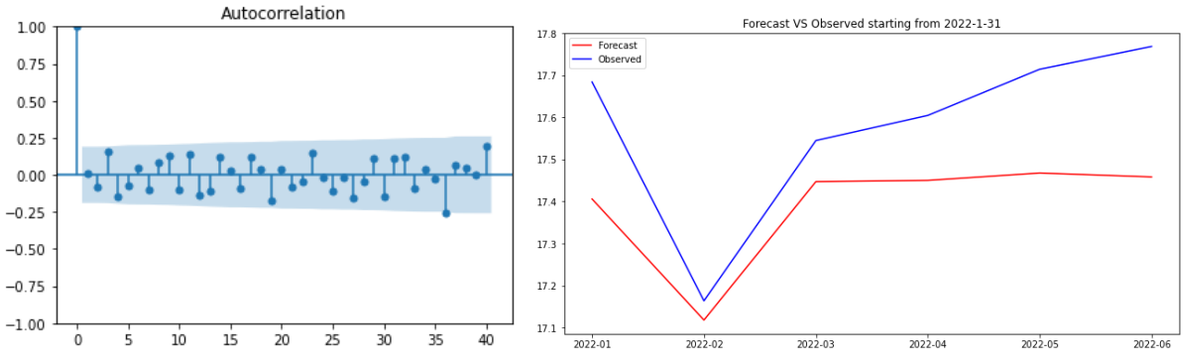


Figure 7: Model forecast: SARIMA(0,1,1)×(2,1,0,12) for ASEAN-EX

3.7 Summary of SARIMA Model Performance

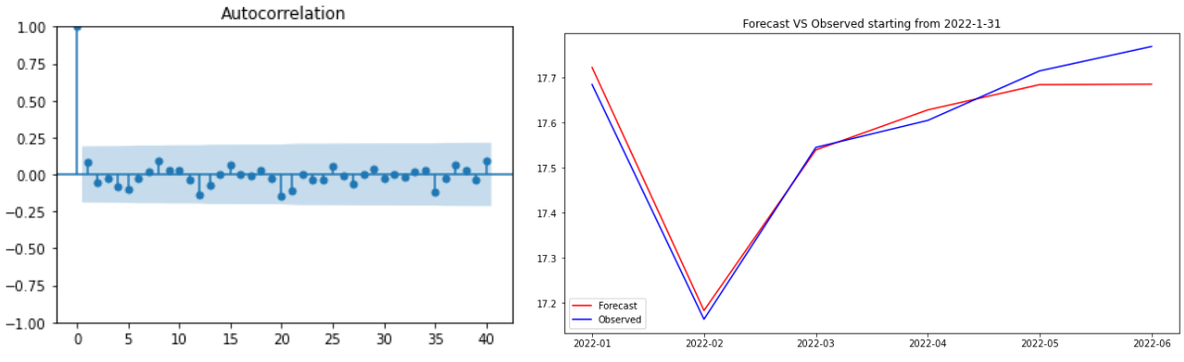
After discussing the ASEAN-EX time series, SARIMA models similar to the above example are fitted into other 5 time series, including ASEAN-IN, EU-EX, EU-IN, APEC-EX, and APEC-IN.

The ASEAN-IN and EU-EX time series are well-fitted with SARIMA (0,1,1)×(2,1,0,12). Figure 8(a) and Figure 9(a) show that the residuals of these two models are approximately white noise. For the 6-month ahead forecast in Figure 8(b) and Figure 9(b), the prediction works well with very low MAEs, which are 0.19 and 0.03 for ASEAN-IN, EU-EX, respectively.



(a) ACF plot of residual (b) Model forecast

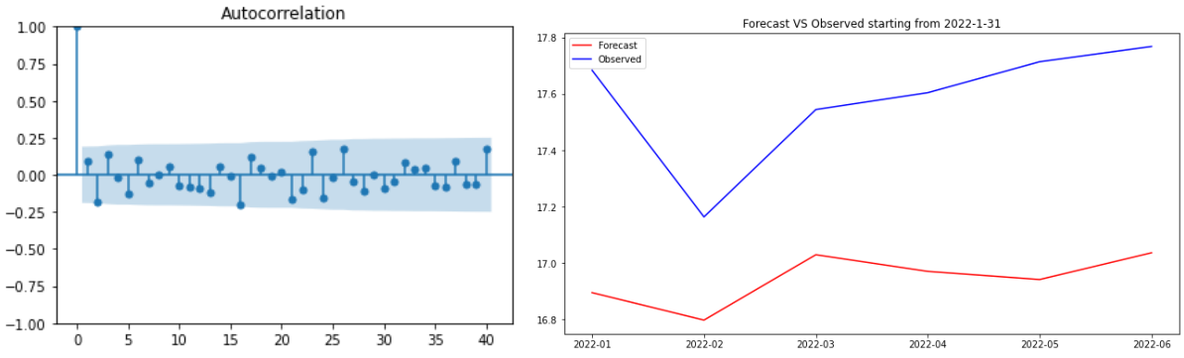
Figure 8: SARIMA (0,1,1)×(2,1,0,12) of ASEAN-IN



(a) ACF plot of residual (b) Model forecast

Figure 9: SARIMA (0,1,1)×(2,1,0,12) of EU-EX

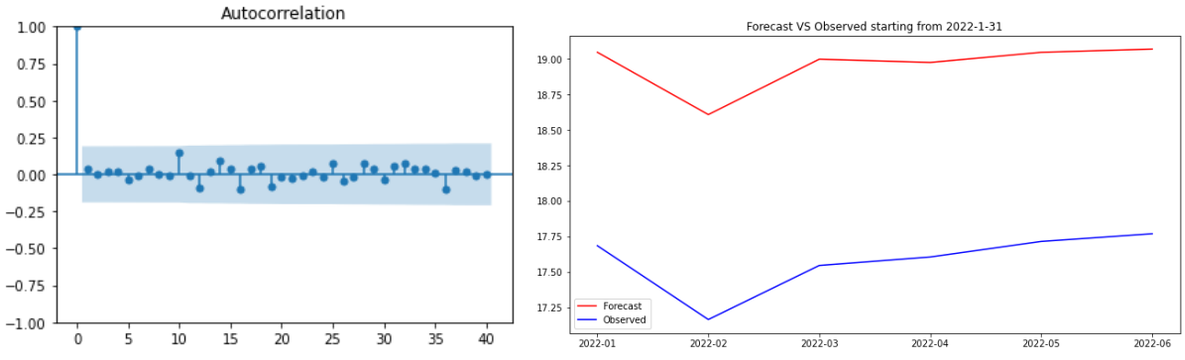
Regarding EU-IN, the performance of SARIMA (0,1,2)×(1,1,0,12) is no longer as good as the first two time series. Although the ACF plot of residual in Figure 10(a) seems to be white noise, we may think SARIMA(0,1,2)×(1,1,0,12) is adequate. However, the gap between our forecast and observation data shown in Figure 10(b) is huge with MAE = 0.64. Even if we try to expand the current SARIMA model, the model will be too large to overfit, but MAE performance for forecasting remains unimproved. Based on the original data's characteristics, we speculate that the change-point may be the cause, which is further discussed in Section 5 Chang-Point Analysis.



(a) ACF plot of residual (b) Model forecast

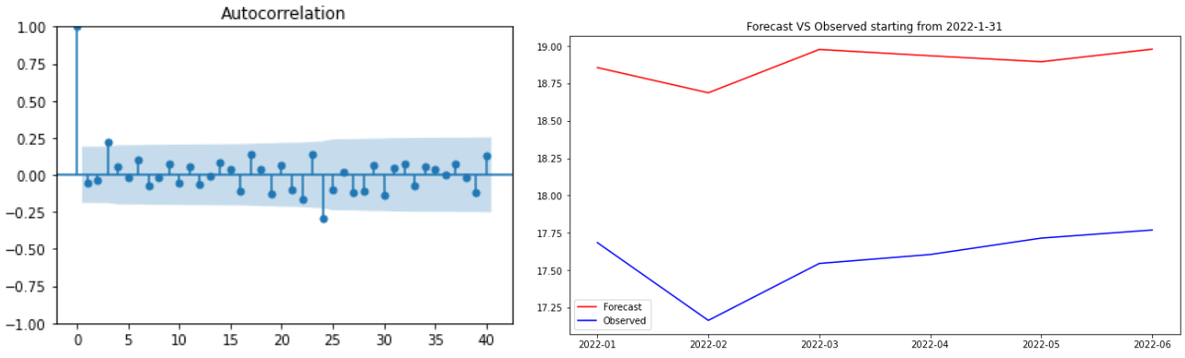
Figure 10: SARIMA (0,1,2)×(1,1,0,12) of EU-IN

Similar problems exist with two APEC-EX and APEC-IN time series. Their SARIMA models cannot properly explain APEC data, as the gap between predicted values and real observations cannot be eliminated if we further expand the model. Also, from Figure 11(b) and Figure 12(b), the MAE performance is poor, with 1.38 and 1.31 for APEC-EX and APEC-IN separately.



(a) ACF plot of residual (b) Model forecast

Figure 11: SARIMA (0,1,1)×(2,1,0,12) of APEC-EX



(a) ACF plot of residual (b) Model forecast

Figure 12: SARIMA (0,1,1)×(1,1,0,12) of APEC-IN

Table 3 summarizes SARIMA models and their performance. To sum up, our Seasonal ARIMA models do fit well for some time series, including ASEAN-EX, ASEAN-IN, and EU-EX. However, the prediction gap can never be eliminated for EU-IN and two time series of APEC. Therefore, other models are considered for a better fit, and one of them is the VAR model to be explained in Section 4 VAR Model.

Time Series	**SARIMA**	**MAE**
ASEAN-EX	(0, 1, 1)×(2, 1, 0, 12)	0.09
ASEAN-IN	(0, 1, 1)×(2, 1, 0, 12)	0.19
EU-EX	(0, 1, 1)×(2, 1, 0, 12)	0.03
EU-IN	(0, 1, 2)×(1, 1, 0, 12)	0.64
APEC-EX	(0, 1, 1)×(2, 1, 0, 12)	1.38
APEC-IN	(0, 1, 1)×(1, 1, 0, 12)	1.31

Table 3: Summary of best SARIMA models for all time series

4 VAR Model

4.1 Model Introduction

After fitting SARIMA models on the six time series data of exports and imports to 3 different regions, our team found that the mean absolute errors were quite large in some cases, such as the APEC export and APEC import. Also, such errors could not converge to 0 when we increase the lag term or seasonal difference. Also, there are cases in the EU where the estimated values were reasonable, but the forecasts performed relatively unsatisfactorily. All these motivated us to seek other additional models or variables that could better explain the variances in the time series and generate better forecasts.

After searching for time series models, we found that the vector autoregression model (VAR) could satisfy our requirement. It can let us include more than one variable in the regression, with all the variables having their lagged terms.[12] Here, we use an example of two variables with one lag.

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix}$$

On the left-hand side, y_1 and y_2 are the variables of interest, and the right-hand side contains constant terms, lagged terms of the variables, and residuals. After expanding the matrix, we have the following equation.

$$y_{1,t} = c_1 + a_{1,1}y_{1,t-1} + a_{1,2}y_{2,t-1}e_{1,t}$$

$$y_{2,t} = c_2 + a_{2,1}y_{1,t-1} + a_{2,2}y_{2,t-1}e_{2,t}$$

In our application, we simplified our understanding to regard the equation as an ordinary linear regression for each dependent variable.

4.2 Variable specification

Since the VAR model can incorporate other variables in the regression, we went further to add other reasonable models that could affect the export or import to a region. First, we determined to add both import and export from the same region to the model. This makes sense because international trading is a reflection of boundedness between two regions and can be affected simultaneously by bilateral relations.

Among the economic indicators, we selected the consumer price index (CPI). Calculated as the average of a basket of domestic goods and services, it is a price index that tracks the price change over time. Unlike another famous economics index, Gross domestic product (GDP),

which is reported quarterly with absolute value, CPI is relatively convenient to get monthly standardized series. Also, since there is a high correlation between various economic factors, adding too many indicators will expand the model but become harder to interpret. Hence, due to parsimony consideration, we restrict our model to only three correlated variables that generously make sense.

4.3 Data processing

We used Python programming language to implement the VAR model. The VAR model requires all the input series to be stationary, whereas Python could not handle this automatically. Hence, based on the insights from SARIMA models, we processed our data manually by common differencing and seasonal differencing. Such operation was also done on the CPI data. Here we take the new CPI series as an example.

Figure 13 is the ACF plot CPI data after taking the logarithm. We noted that although the ACF decreased fast at the beginning, it has a significant seasonal correlation when the lag is 12. Hence, we took one seasonal difference and one common difference as all the time series of export and import.

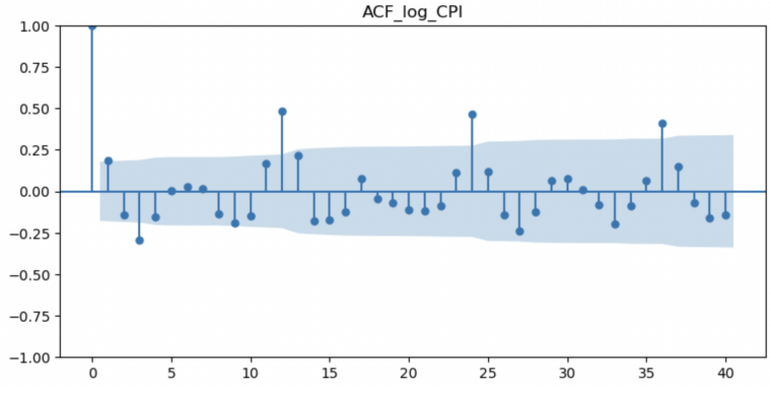


Figure 13: ACF_CPI_LOG

Figure 14 illustrates the ACF plot of log-CPI data with one lag-1 difference and one seasonal difference. We conclude it is a typical time series with one moving average lag and one seasonal moving average lag. Also, we conducted the Dickey-Fuller unit root test on all the time series with the same operation to ensure they all became stationary before fitting into the VAR models.

Figure 15 is the Python output for stationary check, and we are now ready to fit the model with these processed data.

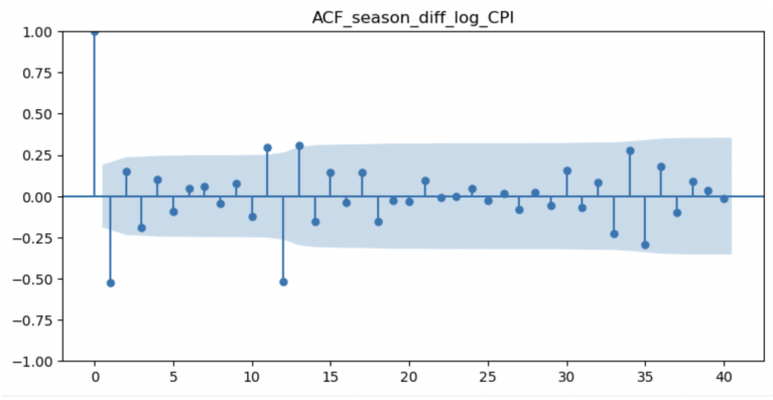


Figure 14: ACF_CPI_DIFF

```
season_diff_log_ASEAN-EX is stationary, continue.
season_diff_log_ASEAN-IN is stationary, continue.
season_diff_log_EU-EX is stationary, continue.
season_diff_log_EU-IN is stationary, continue.
season_diff_log_APEC-EX is stationary, continue.
season_diff_log_APEC-IN is stationary, continue.
season_diff_log_CPI is stationary, continue.
```

Figure 15: stationary check for all series

4.4 VAR model specification

The only parameter that requires specification is the maximum lag. It determines how many lagged terms are included in the model for each variable. There may be many techniques to check the reasonable lagged term, but to simplify the assumption aligned with the SARIMA model, we still use the AIC value for parameter selection.

Table 4 lists the value of AIC with different lagged terms. Here, the AIC varies little with a different number of lags included. To balance the model complexity and explainability, we finalized lag to be 6 in all the VAR models of the three regions.

Table 4: VAR AIC comparison

lag	ASEAN	EU	APEC
3	-18.31	-18.36	-19.22
4	-18.22	-18.46	-19.15
5	-18.48	-18.46	-19.16
6	-18.44	-18.48	-19.20
7	-18.33	-18.38	-19.10
8	-18.24	-18.32	-18.96

lag of CPI	EU-EX	APEC-EX
LAG1.CPI	-6.3 (0.05)	-3.86 (0.103)
LAG2.CPI	0.17 (0.967)	0.35 (0.903)
LAG3.CPI	-7.63 (0.093)	-4.05 (0.195)
LAG4.CPI	-13.17 (0.003)	-6.31 (0.038)
LAG5.CPI	-11.53 (0.006)	-5.79 (0.054)
LAG6.CPI	-6.53 (0.052)	-3.77 (0.121)

Table 5: VAR export CPI correlation

Time Series	SARIMA MAE	VAR MAE
ASEAN-EX	0.09	0.09
ASEAN-IN	0.19	0.13
EU-EX	0.03	0.14
EU-IN	0.64	0.07
APEC-EX	1.38	0.06
APEC-IN	1.31	0.14

Table 6: Comparison summary of best SARIMA models and VAR models

4.5 Result and interpretation

Since all the models contain about 20 explanatory variables, the model output is quite long. Please find the whole result in the appendix from Figure 25 to Figure 29.

We noticed that besides the significant correlation with the lagged terms of the variables themselves, in the model summary of EU export and APEC export, both have shown significant negative relationships with the lagged terms of CPI difference. Table 5 summarizes the coefficient with each p-value in the brackets, and the significant terms are marked as red. It is quite reasonable because CPI is an indicator of domestic inflation, which has a positive impact on the interest rate. When the interest rate increases, the domestic money becomes strong as people are trying to exchange for CNY [9]. In such a case, the export will decrease with a higher price that is not quite attractive to foreign countries. This indicates that the VAR model has successfully captured more variance of the export value besides the lagged variable itself.

4.6 Model forecast

We conducted the same length of forecasts as the SARIMA model and compared their mean absolute error. Table 6 demonstrates that the forecast accuracy has increased a lot in terms of import to the EU and export and import to APEC. Whereas the VAR model did not improve much for those that have already been explained by the variables themselves such as ASEAN-

EX and EU-IN.

Finally, we give the comparison plot in Figure 16 of the predicted value from VAR models and the real observations from 2022 January to 2022 June.

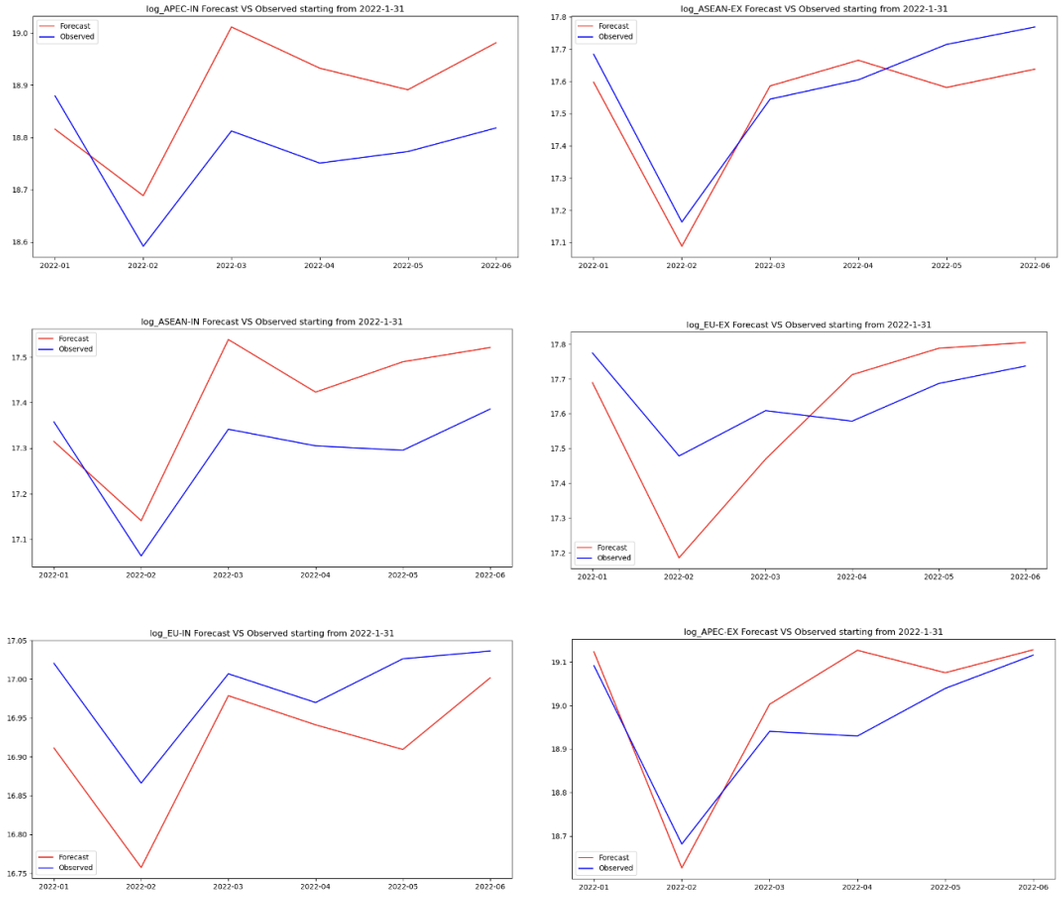


Figure 16: VAR-Forecast vs Obsercation

5 Chang-Point Analysis

5.1 Introduction to Chang-Point Analysis

Sections 3 and 4 investigated the univariate and multivariate modeling to forecast China's imports and exports with ASEAN, EU, and APEC. Imports from ASEAN, exports to ASEAN, and exports to EU countries could be accurately forecasted using customized SARIMA models. Meanwhile, in order to simultaneously model the unexplained variances, we leveraged VAR to incorporate CPI in accompaniment with the import (export) of the same region to forecast the export (import, respectively). VAR outperformed SARIMA models in all time series except exports to EU countries with a significantly lower MAE. A natural question to ask was why incorporating CPI could improve the forecast errors of some time series but not all. Could we identify the missing variables for our target time series prior to conducting multivariate analysis to improve the prediction more efficiently and effectively? The answer was YES, and the technique we employed was called "Change-Point Analysis".

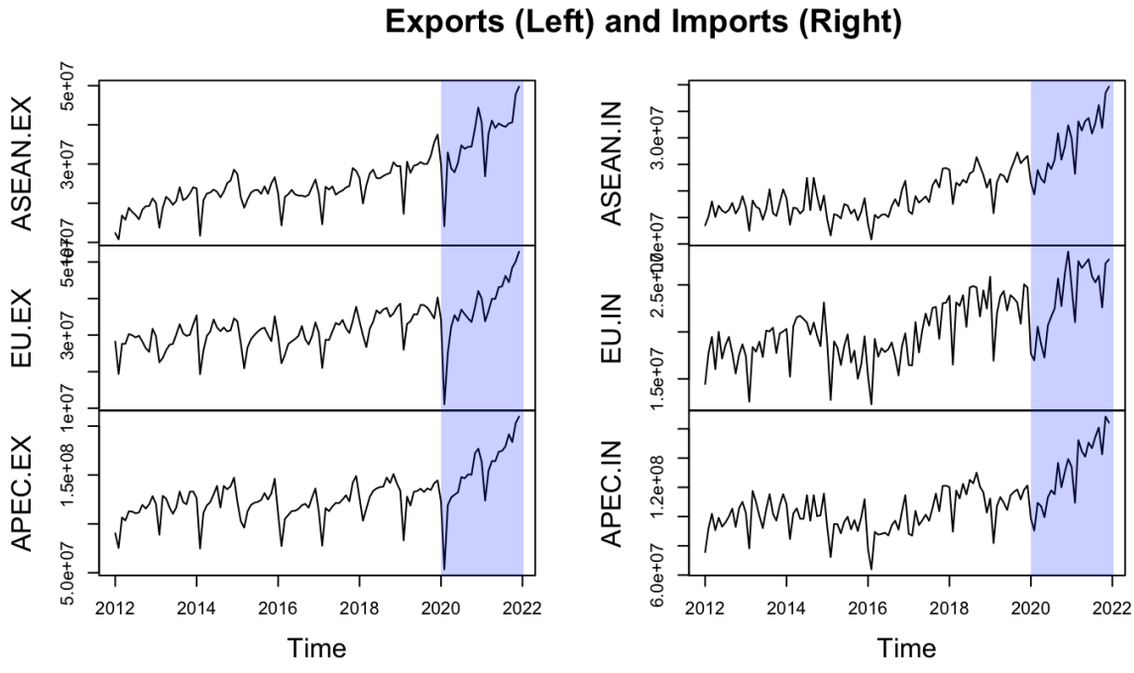


Figure 17: Six time series plots

Before giving a formal definition of change-point analysis, we can observe from the six time series in Figure 17 that the value of imports and exports elevated remarkably after 2020. One of the possible reasons could be the outbreak of COVID-19 in early 2020 globally, which contributed to a surge in world commodity prices. Therefore, we might need to fit two different time series models before and after 2020 or add COVID-19 as an external variable to our model.

Identifying COVID-19 as a possible external factor using our naked eyes is precisely the process of change-point analysis. A change-point, by definition, divides a time series into two segments where each segment has its own statistical characteristics [13], such as mean and variance. Our project focused on the change point in mean because of its economic implication. It is noteworthy that a time series could have one or more change points. The disjoint parts of the time series separated by change points are known as states of the time series or periods of time when the parameters regulating the process remain constant.

Generally, the change-point detection concerns both whether or not one or more changes have occurred and where each change-point is located. Correspondingly, change-point analysis has two primary benefits. First of all, the existence of a change point alerts the data practitioner to re-examine the current model. For instance, the current effective time series model may no longer be effective after an impactful policy change. Secondly, determining the time of the change point eases our identification of possible external factors, which could ideally improve our forecasts when included.

There are many ways to detect change points, which are categorized into offline and online change-point detection. The latter is tailor-made for live streaming data, while the former requires the complete time series for statistical analysis. Hence, we leveraged offline change-point detection techniques in our project because our data is complete. For single change-point detection, we utilized At Most Once Change (AMOC) to detect mean shifts [8]. In terms of detecting multiple change points, we applied Bayesian Change-Point (BCP) analysis methods [14], which outputs a series of estimated posterior means and probabilities of a change at every timestamp.

5.2 Single Change-Point Detection and Analysis

Although we could identify COVID-19 as one of the possible external factors, the real impacts of COVID-19 on different regions were different because of their distinct import and export structures with China. Plus, each time series might have unique external factors due to bilateral trade agreements, wars, geographical reasons, policy changes, and etc. First, to maintain a simple model with the concept of parsimony, we detected a single change point in the mean using AMOC on six time series to identify possible external variables for each of them. In this subsection, we will present the results region by region and hence discuss the possible external variables by delving into the time of the change.

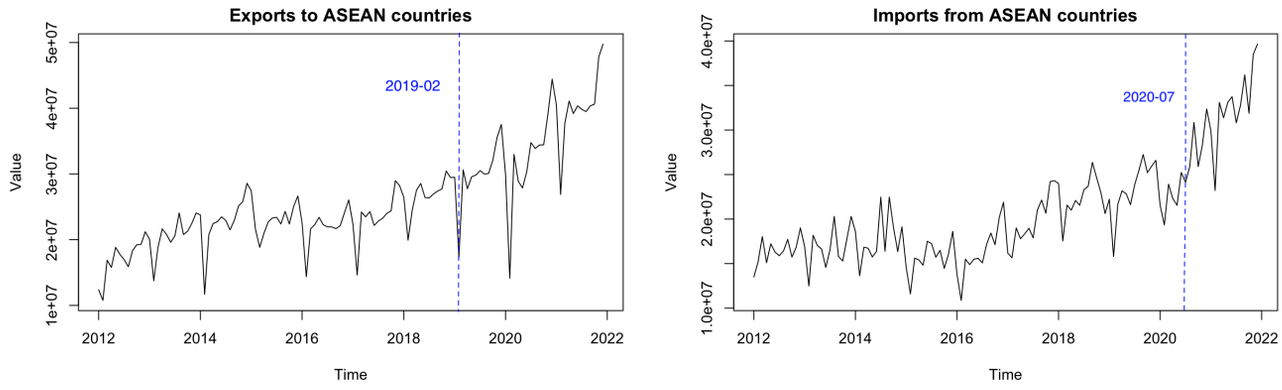


Figure 18: The change-point of ASEAN-EX on the left and ASEAN-IN on the right respectively

As shown in Figure 18, exports to ASEAN have a change point around 2019-02 while imports from ASEAN have a change point around 2020-07. China and ASEAN countries have trade complementarity indicated by the fact that China imports crude oil, coal, and natural rubber from ASEAN to meet its domestic needs [3]. In addition, due to the geographical proximity and good political relations between China and ASEAN nations, bilateral economic and trade relations have been further developed with the reduction of China's tariff after the WTO accession. In 2019, ASEAN–China Free Trade Area (ACFTA) amendments were introduced to ease market access so that the exports to ASEAN countries continued increasing [11]. In Q1 2020, ASEAN nations overtook the EU to become China's largest trading partner because of the EU's stagnation due to COVID-19 and a continuous boost of mechanical and electrical product exports from China to ASEAN nations, further confirming an increasing trend of trading values between China and ASEAN nations. Therefore, China's tariff, COVID-19 statistics, and mechanical and electrical product exports are vital external factors, which could lead to a more accurate and explainable predictive multivariate model.

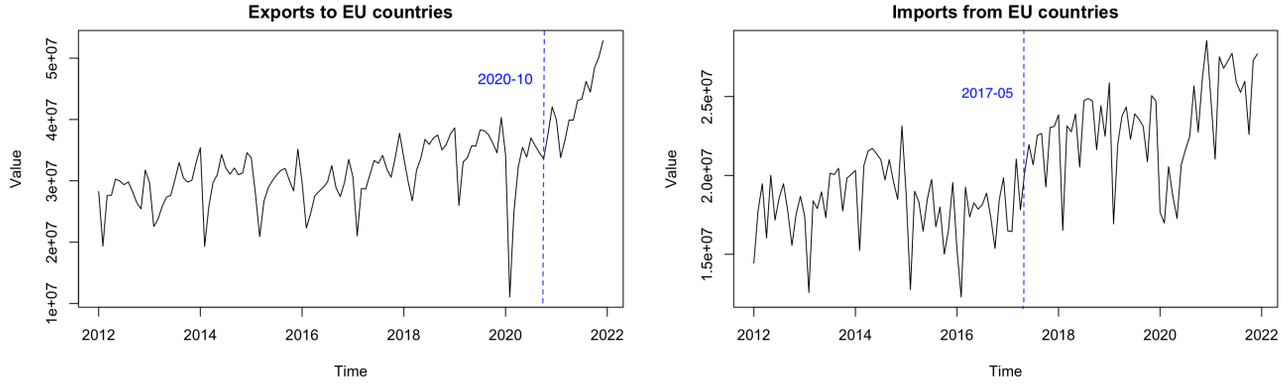


Figure 19: The change-point of EU-EX on the left and EU-IN on the right respectively

From Figure 19, we can observe that the change points of exports to the EU and imports

from the EU are 2020-10 and 2017-05, respectively. The steady increase in exports shall be caused by COVID-19. We can observe the same increase after 2020 for imports from the EU as well. However, the single change-point of EU-IN is in 2017, much earlier than the outbreak of COVID-19, indicating the existence of more external variables. In particular, the Belt and Road Initiative in 2017 promoted imports from EU countries [2]. Accordingly, this initiative might be included in forecasting imports from the EU. By closely examining the EU-IN time series, we can perceive some downturns and upturns e.g. The mean of imports from 2015 to 2017 was steadily lower than in previous periods. Therefore, we suggest multiple change-point detections for EU-IN, which will be discussed in the following subsection.

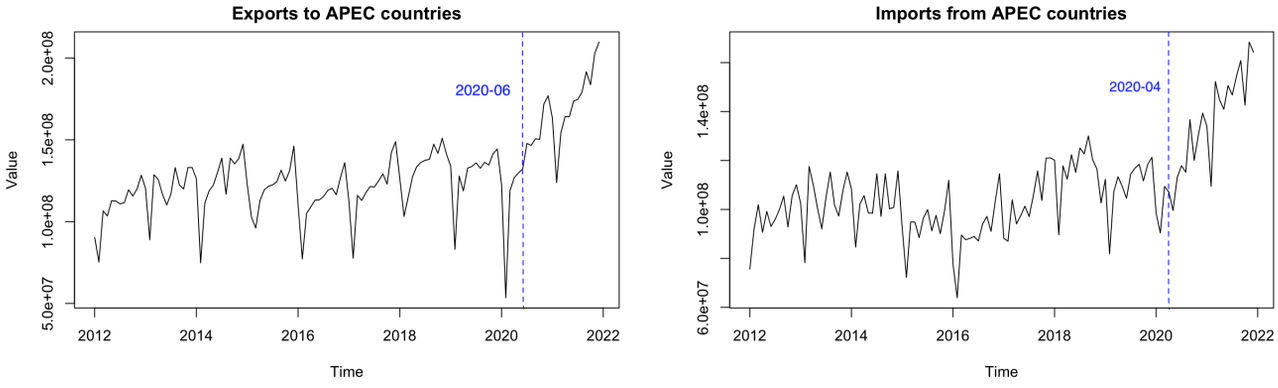


Figure 20: The change-point of APEC-EX on the left and APEC-IN on the right respectively

The single change points of exports to and imports from APEC countries are 2020-06 and 2020-04 respectively as indicated in Figure 20. Through comprehensive research of the events in 2020 between China and APEC countries, COVID-19 is the most significant cause of the surge in imports and exports.

In conclusion, for ASEAN-EX and ASEAN-IN, China's tariff, COVID-19 statistics, and mechanical and electrical exports from China to ASEAN countries are possible external factors. In terms of EU-EX, APEC-EX, and APEC-IN, COVID-19 statistics shall be sufficient to improve the accuracy of forecasts. However, since EU_IN demonstrated more than one change points, we performed multiple change-point detections.

5.3 Multiple Change-Point Detection and Analysis

In this subsection, we leveraged Bayesian Change-Point (BCP) analysis to identify a series of probability of changing points for EU-IN. As shown below in Figure 21, the posterior probability indicates a possible change-point at a location around 40, 62, and 100. These locations correspond to 2015-01, 2017-02, and 2020-02. In 2015, China's imports and exports fell drastically due to the sharp decline of global commodity prices [1]. Then, in 2017, the Belt and Road Initiative motivated trade between the EU and China, boosting the value of imports from EU countries [2]. In early 2020, COVID-19 had a prompt severe negative impact on imports from EU nations, but it soon recovered because China was the only major global economy with growing demand in 2020, especially stoking demand for European cars and luxury goods [5]. Therefore, these three factors shall be considered in modeling to better understand EU-IN and improve forecast accuracy.

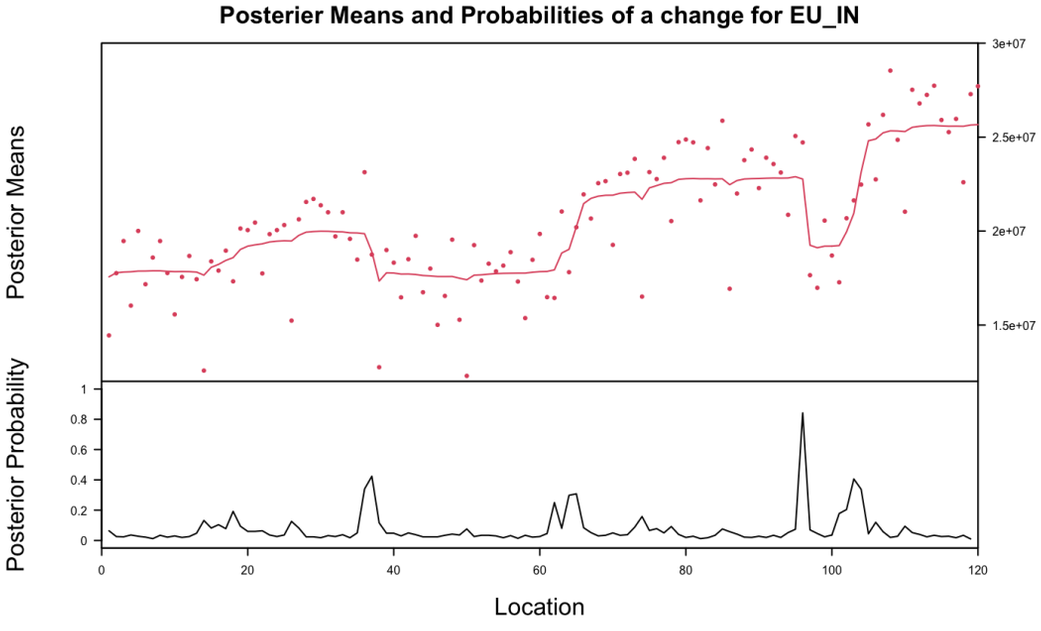


Figure 21: BCP results of EU-IN

To conclude, in this section, we performed a single change-point analysis on six time series and multiple change-point analyses on EU-IN to detect possible external factors to be included in modeling and determine the timeliness of our time series models. With the help of change-point analysis, we could easily and effectively further improve the forecast accuracy by incorporating these external factors into our multi-variate time series models.

6 Conclusion

In this project, we explored six time series of China's imports and exports to the EU, APEC, and ASEAN using data spanning the years 2012 to 2022. We discovered that these series exhibit distinctive trends over time. They reflected ingrained internal patterns, relationships with economic forces, and other external impacts. All of these may vary by location, which illustrates the importance of our study on the import and export data of this critical worldwide economy to various international partners separately.

By examining those data, we conclude that the univariate time series model, SARIMA, can accurately model ASEAN-EX, ASEAN-IN, and EU-EX with a forecast MAE of 0.07, 0.19, and 0.03, respectively. Adding CPI and the imports/exports in the same region to construct the multivariate time series model, VAR, the forecast MAEs of all six time series dropped except for EU-EX, indicating the necessity of customizing external variables for each time series. Therefore, finally, we conducted a single change-point analysis for all time series to determine possible external factors individually by analyzing the import and export structure, policy changes, and major events during the time of change. Furthermore, for EU-IN, which demonstrated multiple change points, we leveraged multiple change-point detection techniques to determine change points. By adding the detected change points as external factors into time series models, our future work is to further reduce the forecast MAE. Hence, more accurate models to forecast China's imports and exports with different regions could facilitate policy-making and budgeting for government officials. With a better means of examining and forecasting data, it is possible for researchers to generalize the stylized facts of the trading system and thus develop new theories to deal with global changes.

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7 Appendix

This part is for massive figures of modeling procedure but not inserted in the body of the report. For the Ljung-box result, the null hypothesis is "the data has no auto-correlation and is regarded as White Noise". A small p-value less than 0.05 indicates the hypothesis is true hence we are not over-differencing in section 3.2 Model specification. Figure 22-24 is the Ljung-box test results of six data sets on 24-time lags respectively. Figure 25 to 30 is the omitted VAR model summary in section 4.5 containing parameter estimation and all related descriptive statistics.

It is not White noise, continue.

	lb_stat	lb_pvalue
1	21.474542	3.585575e-06
2	21.896236	1.759109e-05
3	21.987185	6.563285e-05
4	22.563846	1.547353e-04
5	22.639921	3.954827e-04
6	24.957835	3.476473e-04
7	27.677861	2.514903e-04
8	28.322132	4.167184e-04
9	29.092254	6.252246e-04
10	29.456691	1.050564e-03
11	34.585259	2.899943e-04
12	57.429796	6.610158e-08
13	62.222874	2.096395e-08
14	62.790964	3.779306e-08
15	62.911988	7.910137e-08
16	63.146658	1.531037e-07
17	63.512357	2.735215e-07
18	73.068726	1.357805e-08
19	78.571094	3.274612e-09
20	78.821245	6.217786e-09
21	80.843840	5.832860e-09
22	83.371828	4.480221e-09
23	83.461317	8.658243e-09
24	87.941678	3.150188e-09

(1)

It is not White noise, continue.

	lb_stat	lb_pvalue
1	20.814450	5.059975e-06
2	22.144417	1.553821e-05
3	26.833231	6.380852e-06
4	28.137696	1.169653e-05
5	28.482729	2.928466e-05
6	28.830268	6.550513e-05
7	29.167234	1.348712e-04
8	29.316029	2.790419e-04
9	29.875331	4.606701e-04
10	32.109736	3.839192e-04
11	43.079182	1.052162e-05
12	59.593422	2.677101e-08
13	59.793910	5.718569e-08
14	62.872165	3.656166e-08
15	63.204112	7.036399e-08
16	64.486163	9.028025e-08
17	65.651529	1.192375e-07
18	66.510784	1.743079e-07
19	70.189919	8.552985e-08
20	72.641431	6.708594e-08
21	73.961654	8.032390e-08
22	73.970597	1.547061e-07
23	76.690193	1.081373e-07
24	79.108110	8.433806e-08

(2)

Figure 22: Ljung-box test on $\bigtriangledown^d \bigtriangledown_s^D ln(Z_t)$: ASEAN-EX and ASEAN-IN

It is not White noise, continue.

	lb_stat	lb_pvalue
1	17.960402	0.000023
2	18.363341	0.000103
3	18.364009	0.000370
4	18.440661	0.001012
5	18.462962	0.002419
6	18.593924	0.004907
7	18.599071	0.009541
8	20.621287	0.008224
9	20.999588	0.012652
10	21.132815	0.020186
11	23.475516	0.015135
12	45.930549	0.000007
13	50.039623	0.000003
14	50.043284	0.000006
15	50.921646	0.000009
16	51.104203	0.000015
17	51.462598	0.000025
18	52.390572	0.000033
19	52.449501	0.000057
20	53.216772	0.000076
21	53.227526	0.000128
22	53.235606	0.000211
23	57.876202	0.000077
24	63.837791	0.000018

(3)

It is not White noise, continue.

	lb_stat	lb_pvalue
1	17.036244	3.667309e-05
2	21.133668	2.575623e-05
3	28.063741	3.521851e-06
4	28.734938	8.848909e-06
5	37.202866	5.453972e-07
6	42.210370	1.671017e-07
7	42.538466	4.094862e-07
8	42.966757	8.911821e-07
9	43.005624	2.150489e-06
10	43.168750	4.639146e-06
11	43.652654	8.357458e-06
12	52.367342	5.336251e-07
13	52.998409	9.042515e-07
14	53.793218	1.393492e-06
15	53.799407	2.837529e-06
16	61.056571	3.470427e-07
17	72.606466	7.638285e-09
18	76.364392	3.678252e-09
19	76.554877	7.243448e-09
20	77.780899	9.315445e-09
21	79.108072	1.137540e-08
22	83.574902	4.144379e-09
23	100.948044	9.653970e-12
24	105.619317	3.255439e-12

(4)

Figure 23: Ljung-box test on $\bigtriangledown^d \bigtriangledown_s^D ln(Z_t)$: EU-EX and EU-IN

It is not White noise, continue.

	lb_stat	lb_pvalue
1	23.697705	1.127169e-06
2	23.768564	6.897981e-06
3	23.858439	2.673887e-05
4	24.096527	7.639384e-05
5	24.121067	2.057852e-04
6	25.134388	3.224335e-04
7	26.457862	4.170577e-04
8	26.469769	8.724741e-04
9	27.257305	1.268026e-03
10	28.050089	1.772211e-03
11	30.099761	1.528425e-03
12	47.877614	3.282728e-06
13	52.984225	9.094049e-07
14	52.985231	1.913499e-06
15	53.303946	3.432100e-06
16	54.561727	4.196468e-06
17	54.821115	7.320504e-06
18	58.507704	3.557022e-06
19	60.596576	3.112875e-06
20	60.599609	5.749302e-06
21	60.610562	1.031923e-05
22	60.724454	1.744970e-05
23	66.083379	4.818774e-06
24	77.461080	1.536132e-07

(5)

It is not White noise, continue.

	lb_stat	lb_pvalue
1	23.172330	1.481133e-06
2	23.892967	6.481988e-06
3	26.308635	8.218586e-06
4	26.454804	2.561756e-05
5	27.964428	3.698621e-05
6	28.790679	6.664391e-05
7	28.798426	1.574574e-04
8	29.403423	2.693345e-04
9	29.894076	4.573026e-04
10	30.032025	8.463694e-04
11	32.810287	5.647181e-04
12	41.114025	4.689768e-05
13	42.304506	5.839347e-05
14	42.818988	9.157112e-05
15	42.830459	1.672807e-04
16	45.681841	1.090234e-04
17	48.446803	7.335495e-05
18	48.641491	1.208077e-04
19	51.942479	6.748601e-05
20	54.829916	4.353454e-05
21	55.146520	6.729853e-05
22	58.919844	3.224207e-05
23	69.679420	1.362811e-06
24	76.233671	2.394380e-07

(6)

Figure 24: Ljung-box test on $\bigtriangledown^d \bigtriangledown_s^D ln(Z_t)$: APEC-EX and APEC-IN

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:           Fri, 02, Dec, 2022
Time:                     15:03:59
--------------------------------------------------------------------
No. of Equations:         3.00000    BIC:                   -17.0607
Nobs:                     101.000    HQIC:                  -17.9391
Log likelihood:           563.159    FPE:                9.02757e-09
AIC:                     -18.5366    Det(Omega_mle):     5.38259e-09
--------------------------------------------------------------------
Results for equation season_diff_log_ASEAN-EX
=============================================================================================
                                coefficient       std. error           t-stat            prob
---------------------------------------------------------------------------------------------
const                            -0.004253         0.011991           -0.355           0.723
L1.season_diff_log_ASEAN-EX      -0.657743         0.113604           -5.790           0.000
L1.season_diff_log_ASEAN-IN       0.042079         0.106635            0.395           0.693
L1.season_diff_log_CPI            0.049687         2.493509            0.020           0.984
L2.season_diff_log_ASEAN-EX      -0.434444         0.121905           -3.564           0.000
L2.season_diff_log_ASEAN-IN      -0.099555         0.123990           -0.803           0.422
L2.season_diff_log_CPI            3.411184         3.135546            1.088           0.277
L3.season_diff_log_ASEAN-EX      -0.232704         0.126216           -1.844           0.065
L3.season_diff_log_ASEAN-IN       0.058900         0.128003            0.460           0.645
L3.season_diff_log_CPI           -2.491938         3.474186           -0.717           0.473
L4.season_diff_log_ASEAN-EX      -0.178697         0.125165           -1.428           0.153
L4.season_diff_log_ASEAN-IN       0.306430         0.130033            2.357           0.018
L4.season_diff_log_CPI           -4.559197         3.373822           -1.351           0.177
L5.season_diff_log_ASEAN-EX      -0.118226         0.115094           -1.027           0.304
L5.season_diff_log_ASEAN-IN       0.542394         0.130839            4.146           0.000
L5.season_diff_log_CPI           -5.315059         3.213695           -1.654           0.098
L6.season_diff_log_ASEAN-EX      -0.046752         0.094449           -0.495           0.621
L6.season_diff_log_ASEAN-IN      -0.020658         0.119807           -0.172           0.863
L6.season_diff_log_CPI           -3.572606         2.526399           -1.414           0.157
=============================================================================================
```

Figure 25: VAR model summary: ASEAN-EX

```
Results for equation season_diff_log_ASEAN-IN
=============================================================================================
                                coefficient       std. error           t-stat            prob
---------------------------------------------------------------------------------------------
const                             0.005976         0.012073            0.495           0.621
L1.season_diff_log_ASEAN-EX      -0.029800         0.114378           -0.261           0.794
L1.season_diff_log_ASEAN-IN      -0.610639         0.107361           -5.688           0.000
L1.season_diff_log_CPI            6.031111         2.510493            2.402           0.016
L2.season_diff_log_ASEAN-EX      -0.129785         0.122735           -1.057           0.290
L2.season_diff_log_ASEAN-IN      -0.428987         0.124834           -3.436           0.001
L2.season_diff_log_CPI            5.780879         3.156902            1.831           0.067
L3.season_diff_log_ASEAN-EX      -0.000005         0.127076           -0.000           1.000
L3.season_diff_log_ASEAN-IN      -0.141888         0.128875           -1.101           0.271
L3.season_diff_log_CPI            1.661208         3.497849            0.475           0.635
L4.season_diff_log_ASEAN-EX       0.149563         0.126018            1.187           0.235
L4.season_diff_log_ASEAN-IN      -0.169569         0.130919           -1.295           0.195
L4.season_diff_log_CPI            3.007304         3.396801            0.885           0.376
L5.season_diff_log_ASEAN-EX       0.128235         0.115878            1.107           0.268
L5.season_diff_log_ASEAN-IN      -0.105704         0.131730           -0.802           0.422
L5.season_diff_log_CPI            1.966095         3.235584            0.608           0.543
L6.season_diff_log_ASEAN-EX       0.285918         0.095093            3.007           0.003
L6.season_diff_log_ASEAN-IN      -0.074944         0.120623           -0.621           0.534
L6.season_diff_log_CPI            0.607962         2.543607            0.239           0.811
=============================================================================================
```

Figure 26: VAR model summary: ASEAN-IN

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:           Fri, 02, Dec, 2022
Time:                     15:04:43
--------------------------------------------------------------------
No. of Equations:         3.00000    BIC:                   -17.0066
Nobs:                     101.000    HQIC:                  -17.8850
Log likelihood:           560.428    FPE:                9.52921e-09
AIC:                     -18.4825    Det(Omega_mle):     5.68169e-09
--------------------------------------------------------------------
Results for equation season_diff_log_EU-EX
=============================================================================================
                                coefficient       std. error           t-stat            prob
---------------------------------------------------------------------------------------------
const                             0.007507         0.014693            0.511           0.609
L1.season_diff_log_EU-EX         -0.647429         0.105147           -6.157           0.000
L1.season_diff_log_EU-IN          0.706630         0.159851            4.421           0.000
L1.season_diff_log_CPI           -6.333024         3.241234           -1.954           0.051
L2.season_diff_log_EU-EX         -0.484885         0.120701           -4.017           0.000
L2.season_diff_log_EU-IN          0.654469         0.193216            3.387           0.001
L2.season_diff_log_CPI            0.176874         4.243121            0.042           0.967
L3.season_diff_log_EU-EX         -0.402466         0.128044           -3.143           0.002
L3.season_diff_log_EU-IN          0.334647         0.208356            1.606           0.108
L3.season_diff_log_CPI           -7.631362         4.547225           -1.678           0.093
L4.season_diff_log_EU-EX         -0.508558         0.134564           -3.779           0.000
L4.season_diff_log_EU-IN          0.114784         0.187311            0.613           0.540
L4.season_diff_log_CPI          -13.172413         4.434642           -2.970           0.003
L5.season_diff_log_EU-EX         -0.320436         0.126856           -2.526           0.012
L5.season_diff_log_EU-IN          0.244800         0.174798            1.400           0.161
L5.season_diff_log_CPI          -11.528737         4.214096           -2.736           0.006
L6.season_diff_log_EU-EX         -0.093089         0.100032           -0.931           0.352
L6.season_diff_log_EU-IN          0.257445         0.148474            1.734           0.083
L6.season_diff_log_CPI           -6.537321         3.369875           -1.940           0.052
=============================================================================================
```

Figure 27: VAR model summary: EU-EX

```
Results for equation season_diff_log_EU-IN
=============================================================================================
                                coefficient       std. error           t-stat            prob
---------------------------------------------------------------------------------------------
const                            -0.002437         0.010212           -0.239           0.811
L1.season_diff_log_EU-EX         -0.029744         0.073077           -0.407           0.684
L1.season_diff_log_EU-IN         -0.507366         0.111096           -4.567           0.000
L1.season_diff_log_CPI            5.893781         2.252648            2.616           0.009
L2.season_diff_log_EU-EX          0.039195         0.083887            0.467           0.640
L2.season_diff_log_EU-IN         -0.301870         0.134285           -2.248           0.025
L2.season_diff_log_CPI            4.025967         2.948957            1.365           0.172
L3.season_diff_log_EU-EX          0.106244         0.088990            1.194           0.233
L3.season_diff_log_EU-IN          0.166514         0.144807            1.150           0.250
L3.season_diff_log_CPI            3.339477         3.160308            1.057           0.291
L4.season_diff_log_EU-EX          0.060095         0.093521            0.643           0.520
L4.season_diff_log_EU-IN          0.221418         0.130181            1.701           0.089
L4.season_diff_log_CPI           -1.335764         3.082063           -0.433           0.665
L5.season_diff_log_EU-EX          0.022960         0.088164            0.260           0.795
L5.season_diff_log_EU-IN          0.035822         0.121484            0.295           0.768
L5.season_diff_log_CPI           -0.136301         2.928784           -0.047           0.963
L6.season_diff_log_EU-EX          0.020345         0.069522            0.293           0.770
L6.season_diff_log_EU-IN          0.090475         0.103189            0.877           0.381
L6.season_diff_log_CPI           -1.692108         2.342053           -0.722           0.470
=============================================================================================
```

Figure 28: VAR model summary: EU-IN

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:           Fri, 02, Dec, 2022
Time:                     14:59:01
--------------------------------------------------------------------
No. of Equations:         3.00000    BIC:                   -17.7335
Nobs:                     101.000    HQIC:                  -18.6119
Log likelihood:           597.135    FPE:                4.60663e-09
AIC:                     -19.2094    Det(Omega_mle):     2.74665e-09
--------------------------------------------------------------------
Results for equation season_diff_log_APEC-EX
=============================================================================================
                                coefficient       std. error           t-stat            prob
---------------------------------------------------------------------------------------------
const                            -0.000021         0.011475           -0.002           0.999
L1.season_diff_log_APEC-EX       -0.659743         0.111293           -5.928           0.000
L1.season_diff_log_APEC-IN        0.239542         0.142474            1.681           0.093
L1.season_diff_log_CPI           -3.860590         2.365953           -1.632           0.103
L2.season_diff_log_APEC-EX       -0.423297         0.127709           -3.315           0.001
L2.season_diff_log_APEC-IN        0.147021         0.173602            0.847           0.397
L2.season_diff_log_CPI            0.354624         2.909717            0.122           0.903
L3.season_diff_log_APEC-EX       -0.200998         0.135414           -1.484           0.138
L3.season_diff_log_APEC-IN        0.077766         0.181972            0.427           0.669
L3.season_diff_log_CPI           -4.053087         3.127383           -1.296           0.195
L4.season_diff_log_APEC-EX       -0.190685         0.134951           -1.413           0.158
L4.season_diff_log_APEC-IN        0.134913         0.179431            0.752           0.452
L4.season_diff_log_CPI           -6.315992         3.039660           -2.078           0.038
L5.season_diff_log_APEC-EX       -0.147694         0.124159           -1.190           0.234
L5.season_diff_log_APEC-IN        0.475822         0.167511            2.841           0.005
L5.season_diff_log_CPI           -5.788710         3.007116           -1.925           0.054
L6.season_diff_log_APEC-EX       -0.083600         0.097789           -0.855           0.393
L6.season_diff_log_APEC-IN        0.084453         0.141666            0.596           0.551
L6.season_diff_log_CPI           -3.773122         2.434487           -1.550           0.121
=============================================================================================
```

Figure 29: VAR model summary: APEC-EX

```
Results for equation season_diff_log_APEC-IN
=============================================================================================
                                coefficient       std. error           t-stat            prob
---------------------------------------------------------------------------------------------
const                             0.002230         0.008997            0.248           0.804
L1.season_diff_log_APEC-EX        0.074476         0.087259            0.853           0.393
L1.season_diff_log_APEC-IN       -0.644774         0.111708           -5.772           0.000
L1.season_diff_log_CPI            4.696939         1.855036            2.532           0.011
L2.season_diff_log_APEC-EX        0.096200         0.100131            0.961           0.337
L2.season_diff_log_APEC-IN       -0.440461         0.136114           -3.236           0.001
L2.season_diff_log_CPI            3.624993         2.281376            1.589           0.112
L3.season_diff_log_APEC-EX        0.152587         0.106172            1.437           0.151
L3.season_diff_log_APEC-IN       -0.180866         0.142676           -1.268           0.205
L3.season_diff_log_CPI            2.136039         2.452038            0.871           0.384
L4.season_diff_log_APEC-EX        0.058192         0.105809            0.550           0.582
L4.season_diff_log_APEC-IN       -0.016379         0.140684           -0.116           0.907
L4.season_diff_log_CPI            2.877601         2.383258            1.207           0.227
L5.season_diff_log_APEC-EX        0.111922         0.097347            1.150           0.250
L5.season_diff_log_APEC-IN        0.017310         0.131338            0.132           0.895
L5.season_diff_log_CPI            2.379920         2.357742            1.009           0.313
L6.season_diff_log_APEC-EX        0.162690         0.076672            2.122           0.034
L6.season_diff_log_APEC-IN        0.036533         0.111074            0.329           0.742
L6.season_diff_log_CPI            2.581772         1.908770            1.353           0.176
=============================================================================================
```

Figure 30: VAR model summary: APEC-IN